

Magnetic Capsule Maze Cup

Steer a magnetic pill-camera through a body maze without touching it.

Win condition: Fastest clean run with low penalties and a clear magnetic-control explanation.

THE SCIENCE YOU ARE USING

Remote actuation. A magnet pulls a steel object through a barrier without contact. Doctors use the same idea to steer a swallowed capsule camera through the gut from outside the body. The field passes through the wall; only the magnetic force does the work.

Path planning. Move too fast and the capsule overshoots; move too slow and you waste time. You plan a route through the maze and control the external magnet to get a fast, clean run with the fewest wall touches.

HOW TO BUILD AND RUN IT

- 1 Test the field through a barrier.** Place the steel token on the maze and move a magnet under the board. Feel how the field pulls through.
- 2 Plan a route.** Trace the maze path you will follow and mark tricky corners.
- 3 Practice control.** Practice slow, steady moves. Sudden jerks make the token jump walls.
- 4 Run for time.** Make a timed clean run. Each wall touch is a penalty.
- 5 Explain the control.** Describe how distance and speed of the magnet changed your control.

Safety: Magnet safety is strict. Keep magnets away from mouths, electronics, and medical implants. Count magnets in and out. Allergy: None.

MATERIALS

Small neodymium magnets	controlled count
Maze boards	per team
Steel tokens or paper clips	per team
Timers	per team
Barrier sheets	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why):

The ONE variable we are testing:

Baseline result:

After redesign:

DEFEND YOUR DESIGN

What evidence made you change your design?

If you ran it again, what is the ONE thing you would change?

HOW YOU SCORE (100 POINTS)

Scoring criterion	Points
Fastest clean run	50
Low penalties	20
Magnetic-control explanation	20
Redesign	10
Total	100